PROJECT REPORT

Laptop Price Prediction Using Machine Learning

Regression and price-category classification on the Kaggle Laptop Price Dataset

Student Name	BHOLA SANKAR MAJHI
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Abstract

Laptop prices vary widely with brand, memory, processor, graphics and other specifications. This project uses the Kaggle Laptop Price Dataset, which contains 1,303 laptops described by 13 columns, to study that variation and to build models that predict price. After a data-quality check and cleaning of the **Ram** and **Weight** columns, exploratory data analysis (EDA) examined the price distribution, brand averages, RAM, laptop type, screen size and feature correlations. Prices are right-skewed, RAM has the strongest linear correlation with price among the numeric inputs (0.743), and screen size has almost none (0.068).

Three regression models were trained to predict price in euros and compared on a held-out test set of 261 laptops. **Linear Regression** achieved the lowest RMSE (€282.91) and the highest R^2 (0.8424); **Random Forest** achieved the lowest MAE (€184.01); and **Gradient Boosting** reached an R^2 of 0.8342. A separate **Logistic Regression** classifier assigned laptops to Low (≤ €700), Medium (€700–€1500) or High (> €1500) price categories, with an accuracy of 0.8506, a macro ROC-AUC of 0.9606 and a weighted ROC-AUC of 0.9581.

Linear Regression is preferred for price prediction on RMSE and R^2, with Random Forest a strong alternative when MAE is the priority. These are test-set estimates from a single train-test split and are not guarantees of future accuracy. Limitations include the modest dataset size, historical prices, no cross-validation or tuning, fixed category thresholds and no external validation.

1. Introduction

A laptop’s price reflects a combination of factors, including the manufacturer, processor, memory, storage, graphics card, display and build type. For buyers, it is hard to judge from a specification sheet alone whether a price is reasonable. For sellers and analysts, understanding how specifications relate to price supports comparison and market analysis. **Laptop price prediction** is the task of estimating a laptop’s price from its specifications using a model trained on laptops with known prices.

This project studies that task in two forms. As a **regression** problem, the model predicts an exact numerical price in euros. As a **classification** problem, the model predicts whether a laptop falls into a Low, Medium or High price category.

Scope. The work is limited to the supplied dataset and the features it contains. It covers data checking and cleaning, EDA, three regression models, one classification model and their evaluation. Predicted prices are estimates based on this historical dataset and should not be read as current market prices.

2. Project Objectives

- Understand and clean the laptop dataset.
- Carry out exploratory data analysis (EDA) and study how specifications relate to price.
- Build a Linear Regression model for price prediction.
- Train and compare Random Forest and Gradient Boosting regressors.
- Evaluate the regression models using MAE, RMSE and R^2.
- Apply Logistic Regression to classify laptops into Low, Medium and High price categories.
- Evaluate classification using accuracy, precision, recall, F1-score and ROC-AUC, and examine a confusion matrix.
- Select suitable models from the results, and identify limitations and future improvements.

3. Dataset Description

The data comes from the Kaggle Laptop Price Dataset (link) and was loaded from the file `laptop_price.csv`. According to the notebook output, it contains **1,303 records** and **13 columns**. Each record is one laptop, and prices are given in euros.

Table 1. Columns of the dataset (types and distinct-value counts from the notebook output).

Column	Type	Description	Distinct values
`laptop_ID`	int64	Unique laptop identifier (dropped before modelling)	1,303
`Company`	object	Laptop manufacturer	19
`Product`	object	Laptop model name	618
`TypeName`	object	Laptop category (e.g. Notebook, Gaming, Ultrabook)	6
`Inches`	float64	Screen size in inches	18
`ScreenResolution`	object	Display resolution and panel description	40
`Cpu`	object	Processor description	118
`Ram`	object → int	Installed RAM (e.g. “8GB”; converted to a number)	9
`Memory`	object	Storage type and capacity	39
`Gpu`	object	Graphics processor	110
`OpSys`	object	Operating system	9
`Weight`	object → float	Weight (e.g. “1.37kg”; converted to a number)	179
`Price_euros`	float64	Laptop price in euros (regression target)	791

Targets. The **regression target** is **Price_euros**. The **classification target** is a price category derived from the same column: **Low** (price ≤ €700), **Medium** (€700 < price ≤ €1500) or **High** (price > €1500).

Table 2. Summary statistics of Price_euros (1,303 laptops).

Statistic	Value (€)
Mean	1,123.69
Standard deviation	699.01
Minimum	174.00
25th percentile	599.00
Median	977.00
75th percentile	1,487.88
Maximum	6,099.00

The most common values reported by the notebook were Dell (297 laptops) for **Company**, Notebook (727) for **TypeName**, 8GB (619) for **Ram**, 256GB SSD (412) for **Memory**, Intel HD Graphics 620 (281) for **Gpu** and Windows 10 (1,072) for **OpSys**.

4. Tools and Libraries

The notebook was developed in **Python** on **Google Colab** (the data file is read from the Colab path /content/). The libraries actually imported and used are listed below. The versions of Python and the libraries are not recorded in the notebook.

Table 3. Python libraries used in the notebook.

Library	Used for
Pandas	Loading the CSV, inspecting data, group-by averages, cleaning columns, creating price categories
NumPy	Numerical operations, such as the square root in RMSE and infinite upper bound for price bins
Matplotlib	Figures and plot formatting
Seaborn	Histograms, bar charts, scatter plots and heatmaps
Scikit-learn	**train_test_split**; **ColumnTransformer**, **Pipeline**, **SimpleImputer**, **OneHotEncoder**; **LinearRegression**, **RandomForestRegressor**, **GradientBoostingRegressor**, **LogisticRegression**; metrics (**mean_absolute_error**, **mean_squared_error**, **r2_score**, **accuracy_score**, **classification_report**, **confusion_matrix**, **roc_curve**, **auc**, **roc_auc_score**); **label_binarize**; **clone**

5. Data Loading and Understanding

The CSV file was loaded with **pandas.read_csv** using the `latin-1` encoding. The notebook then inspected the shape, the first and last five records, column names, data types, a summary of statistics for every column and the number of unique values per column.

The loaded data has the shape (1,303, 13). Ten columns are stored as text (**object**), two as floating-point numbers (**Inches**, **Price_euros**) and one as an integer (**laptop_ID**). All 13 columns have 1,303 non-null values. Screen sizes range from 10.1 to 18.4 inches, with a mean of about 15.02.

5.1 Data quality checks

Table 4. Data-quality checks performed in the notebook.

Check	Result
Missing values (all columns)	0
Duplicate rows	0
Duplicate laptop_ID values	0
Prices ≤ 0	0
Screen sizes ≤ 0	0

The data is complete and contains no duplicate records or invalid prices or screen sizes. The one issue found is that **Ram** and **Weight** hold numbers as text, which is dealt with in the next section. The checks cover missing values, duplicates and impossible values only. The notebook did not check for outliers or for errors in the text columns (such as inconsistent product names).

6. Data Cleaning and Preprocessing

6.1 Cleaning operations

- A copy of the data, **df_clean**, was created so the original stays unchanged.
- **Ram**: the suffix “GB” was removed and the result was converted to a number. The cleaned values are 2, 4, 6, 8, 12, 16, 24, 32 and 64.
- **Weight**: the suffix “kg” was removed and the result was converted to a decimal number.
- Both conversions created no missing values, and the shape remained (1,303, 13).

6.2 Preparing features for modelling

- **laptop_ID** was dropped, because it is an identifier and not a laptop specification.
- **Price_euros** was separated as the target (**y**). The remaining **11 columns** form the input features (**X**).
- **Numerical features (3):** Inches, Ram, Weight.
- **Categorical features (8):** Company, Product, TypeName, ScreenResolution, Cpu, Memory, Gpu, OpSys.

6.3 Preprocessing pipeline

Preprocessing was built with scikit-learn's **ColumnTransformer** and combined with each model in a single **Pipeline**, so that all preprocessing steps are fitted on the training data only. Numerical columns pass through a median **SimpleImputer**. Categorical columns pass through a most-frequent **SimpleImputer** followed by **OneHotEncoder** with handle_unknown set to "ignore". The dataset has no missing values, so the imputers do not change any value here; they only make the pipeline robust. No feature scaling, outlier removal or new features derived from the text columns (for example, extracting processor speed or storage size) were applied, and each text column was one-hot encoded as it appears in the data.

6.4 Train-test splitting

Table 5. Train-test splits used in the notebook.

Task	Split	Training rows	Test rows	Notes
Regression (all three models)	80% / 20%	1,042	261	random_state = 42
Classification (Logistic Regression)	80% / 20%	1,042	261	random_state = 42, stratified by price category

All three regression models were trained and tested on the same split, so their results are directly comparable. The classification split is stratified, which keeps the category proportions similar in the training and test sets. It therefore contains a different set of 261 laptops from the regression test set.

Note: **Product** has 618 distinct names, so its one-hot encoding adds a very large number of sparse columns. The notebook did not test the effect of removing this feature, so its influence on the results cannot be judged from the notebook.

7. Exploratory Data Analysis (EDA)

EDA was used to understand the price range and to see which specifications are associated with price. All observations below describe associations within this dataset; none of them shows that one feature causes a higher price.

7.1 Price distribution

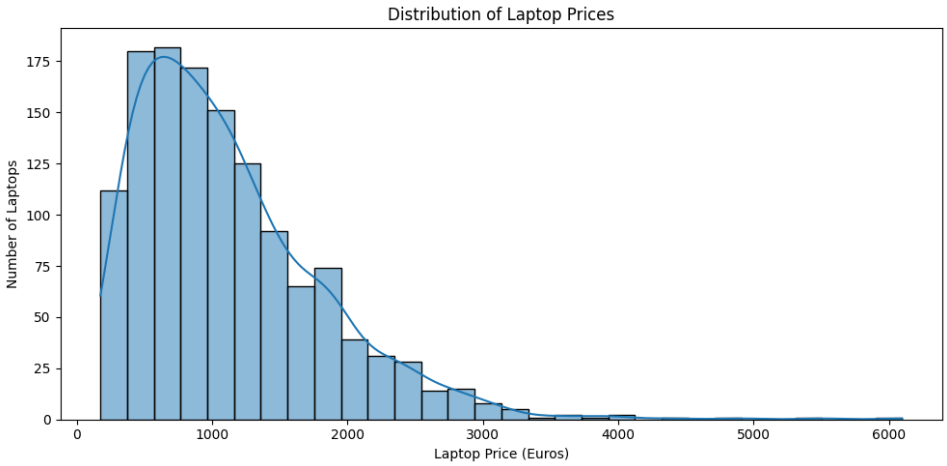


Figure 1. Distribution of laptop prices (histogram with density curve).

Most laptops are priced between roughly €500 and €1,500. The count of laptops falls as price rises, giving a **right-skewed** distribution with a long tail towards high prices. The mean (€1,123.69) is higher than the median (€977.00), which is consistent with that skew. A small number of laptops cost more than about €3,000, and a few reach roughly €6,000 (the maximum is €6,099). Such expensive laptops are likely to be harder for a model to predict, because there are only a few of them.

7.2 Brand comparison

Table 6. Average laptop price by brand (19 brands), from highest to lowest.

Brand	Avg. price (€)	Brand	Avg. price (€)
Razer	3,346.14	Xiaomi	1,133.46
LG	2,099.00	Asus	1,104.17
MSI	1,728.91	Lenovo	1,086.38
Google	1,677.67	HP	1,067.77
Microsoft	1,612.31	Fujitsu	729.00
Apple	1,564.20	Acer	626.78
Huawei	1,424.00	Chuwi	314.30
Samsung	1,413.44	Mediacom	295.00
Toshiba	1,267.81	Vero	217.42
Dell	1,186.07		

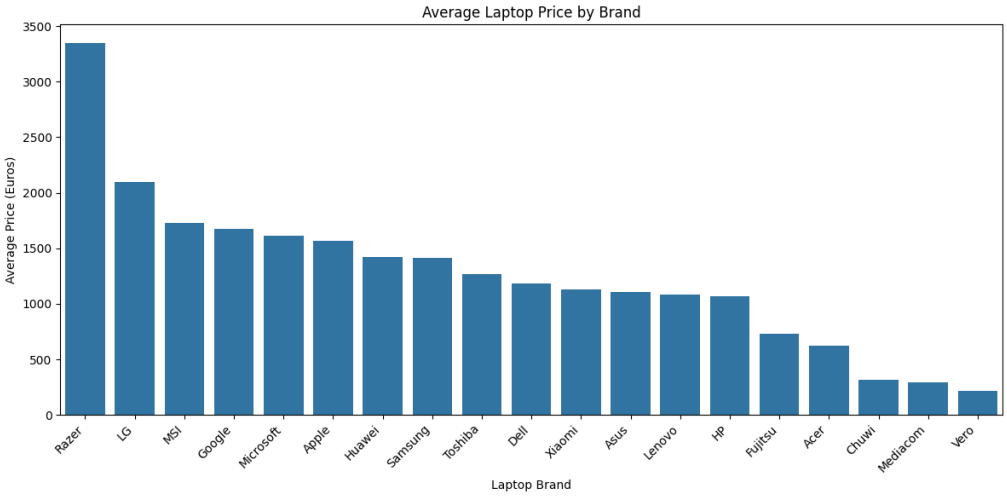


Figure 2. Average laptop price by brand.

Average price differs considerably between brands. **Razer** is highest (about €3,346.14), followed by **LG** (€2,099.00) and **MSI** (€1,728.91); **Apple** averages €1,564.20. **Vero** is lowest (€217.42), with **Mediacom** and **Chuwi** also low. These are averages over the models that each brand has in this dataset. The notebook does not show how many laptops each brand contributes, so brands with few models may have unstable averages. A high brand average also does not mean every laptop from that brand is expensive.

7.3 RAM and price

Table 7. Average price and number of laptops by RAM size.

RAM (GB)	Avg. price (€)	Laptops
2	254.37	22
4	574.57	375
6	615.21	41
8	1,180.81	619
12	1,239.44	25
16	1,936.77	200
24	2,206.33	3
32	3,413.09	17
64	3,975.00	1

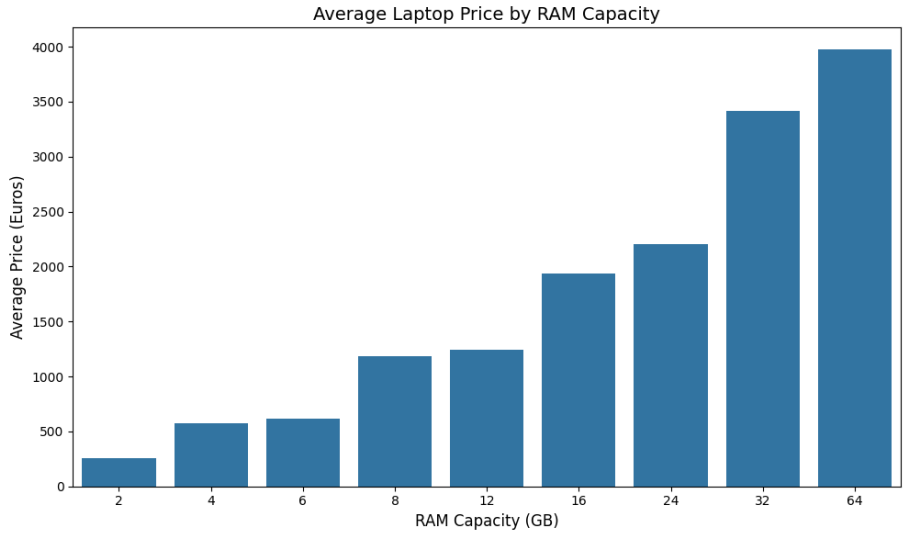


Figure 3. Average laptop price by RAM capacity.

Average price generally rises with RAM, with a noticeable jump between 8 GB (€1,180.81) and 16 GB (€1,936.77). The 8 GB group is the largest (619 laptops). The 24 GB group (3 laptops) and 64 GB group (1 laptop) are very small, so their averages are not reliable. RAM is associated with price here, but laptops with more RAM also tend to differ in brand, processor, storage and graphics, so the figures do not show that RAM alone raises the price.

7.4 Laptop type and price

Table 8. Average price and number of laptops by type.

Laptop type	Avg. price (€)	Laptops
Workstation	2,280.36	29
Gaming	1,731.38	205
Ultrabook	1,548.23	196
2 in 1 Convertible	1,282.40	121
Notebook	782.08	727
Netbook	636.23	25

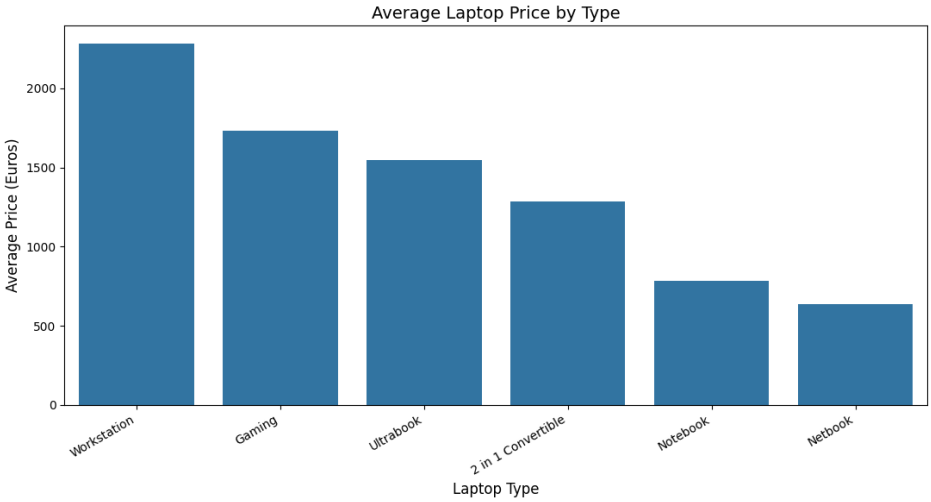


Figure 4. Average laptop price by laptop type.

Workstations have the highest average price (€2,280.36) and Netbooks the lowest (€636.23). Gaming laptops and Ultrabooks average more than Notebooks. Notebooks are by far the largest

group (727 of 1,303 laptops), while Workstation and Netbook each have fewer than 30 laptops, so differences involving those groups should be read with care.

7.5 Screen size and price

Table 9. The four most common screen sizes (together 1,190 of 1,303 laptops).

Screen size (in)	13.3	14.0	15.6	17.3
Avg. price (€)	1,332.32	1,202.27	903.36	1,673.06
Laptops	164	197	665	164

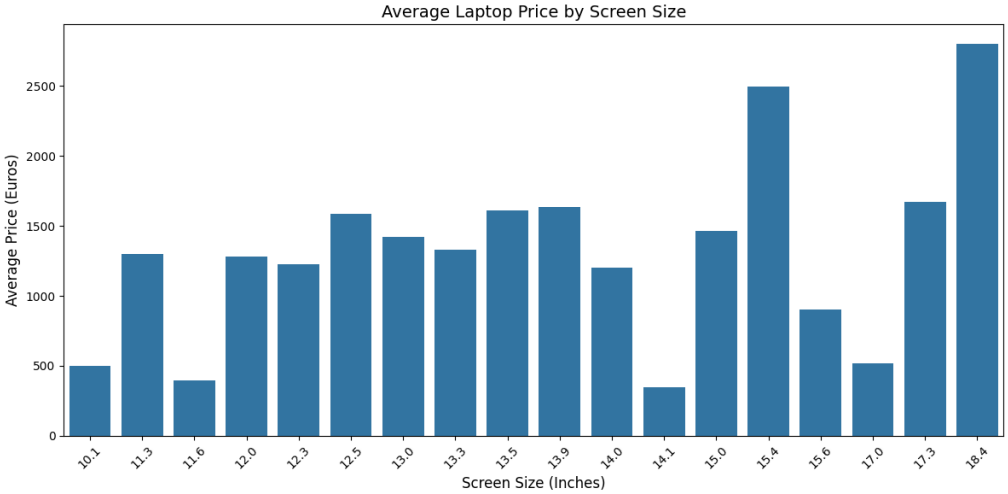


Figure 5. Average laptop price by screen size.

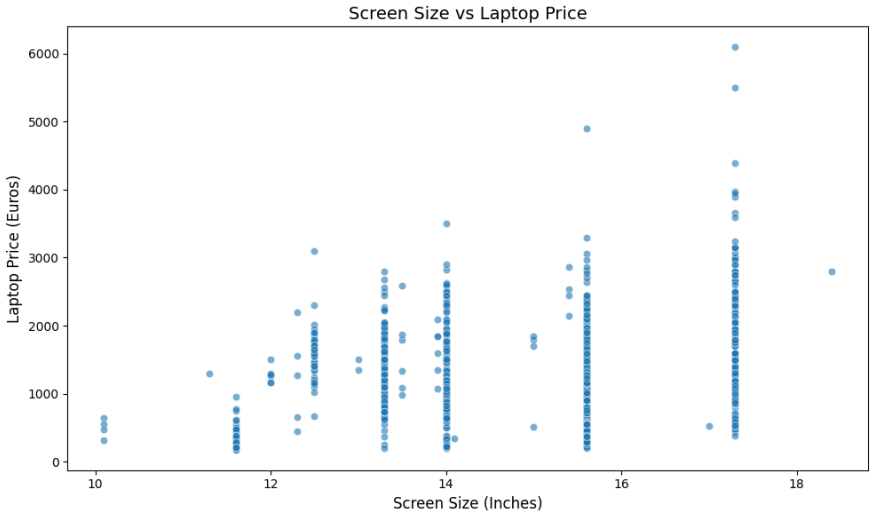


Figure 6. Screen size against laptop price (each point is one laptop).

Screen sizes range from 10.1 to 18.4 inches, and the 15.6-inch size is by far the most common (665 laptops). Average price does not rise steadily with screen size. For example, 15.4-inch laptops average €2,493.85 while 15.6-inch laptops average €903.36, although several sizes have only one to six laptops. The scatter plot shows a wide spread of prices at similar screen sizes. The correlation between screen size and price is only **0.068**, a very weak positive linear relationship.

7.6 Correlation of numerical features

Table 10. Correlation matrix of the numerical features and price.

	Inches	Ram	Weight	Price_euros
Inches	1.000	0.238	0.828	0.068
Ram	0.238	1.000	0.384	0.743
Weight	0.828	0.384	1.000	0.210
Price_euros	0.068	0.743	0.210	1.000

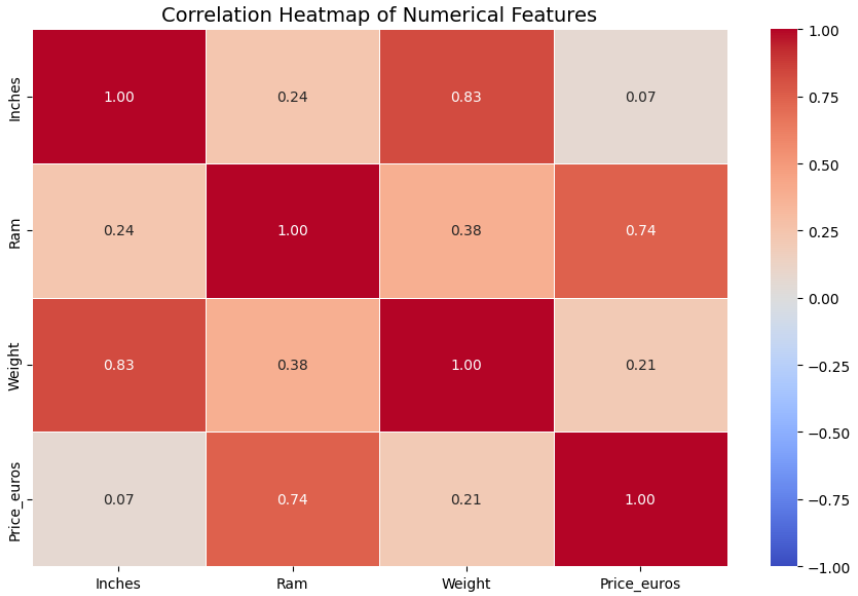


Figure 7. Correlation heatmap of the numerical features.

RAM has the strongest linear correlation with price (0.743), followed by weight (0.210) and screen size (0.068). Screen size and weight are strongly correlated with each other (0.828). Correlation measures linear association only and does not prove causation. A weak correlation does not make a feature useless to a model, because the relationship may be non-linear or may depend on other features. The heatmap covers only three numerical features, so it says nothing about categorical features such as brand, processor, graphics or laptop type.

8. Machine Learning Methodology

Regression vs classification. Regression predicts a **number**, here the price in euros, and is judged by how far the predictions are from the true prices (MAE, RMSE, R^2). Classification predicts a **category**, here Low, Medium or High, and is judged by how often and how well the categories are identified (accuracy, precision, recall, F1-score, ROC-AUC). Because the metrics measure different things, regression and classification results are not directly comparable.

8.1 Linear Regression

Linear Regression models the price as a weighted sum of the input features, with the weights chosen to minimise the squared difference between predicted and actual prices on the training data. It is simple and fast, and it gives a useful baseline. It can only capture linear relationships between the (encoded) features and price. Default settings were used.

8.2 Random Forest Regressor

A Random Forest is an ensemble of many decision trees, each trained on a random sample of the data. The final prediction is the average of the trees' predictions. Averaging many trees reduces the risk of overfitting to the training data, and trees can capture non-linear patterns and interactions. Settings used: 100 trees, random_state = 42, n_jobs = −1, with other settings at their defaults.

8.3 Gradient Boosting Regressor

Gradient Boosting also uses decision trees, but it builds them one after another. Each new tree tries to correct the errors left by the trees before it. Settings used: 100 estimators, learning rate 0.1, maximum depth 3 and random_state = 42.

8.4 Logistic Regression (price-category classification)

Despite its name, Logistic Regression is a **classification** method. For each laptop it estimates the probability of each price category and predicts the category with the highest probability. Here the three categories are Low (price ≤ €700), Medium (€700 < price ≤ €1500) and High (price > €1500). The categories were created from **Price_euros** with **pandas.cut**. The model was run with max_iter = 2000 and the same preprocessing pipeline as the regression models.

Table 11. Models used and how each is evaluated.

Model	Task	Output	Evaluated with
Linear Regression	Regression	Price in euros	MAE, RMSE, R^2
Random Forest	Regression	Price in euros	MAE, RMSE, R^2
Gradient Boosting	Regression	Price in euros	MAE, RMSE, R^2
Logistic Regression	Classification	Low / Medium / High	Accuracy, precision, recall, F1, ROC-AUC

9. Regression Model Evaluation

9.1 Evaluation metrics

- **MAE (Mean Absolute Error)** is the average size of the prediction errors, ignoring direction: mean of |actual − predicted|. It is in euros and is easy to interpret. Lower is better.
- **RMSE (Root Mean Squared Error)** is the square root of the mean of the squared errors. It is also in euros, but squaring gives large errors extra weight, so RMSE is higher than MAE when a few predictions are far off. Lower is better.
- **R^2 (coefficient of determination)** is the share of the variation in test prices that the model explains, compared with always predicting the mean price. A value of 1 is a perfect fit. Higher is better.

9.2 Model comparison

Table 12. Regression results on the 261-laptop test set. The best value in each column is shaded.

Model	MAE (€)	RMSE (€)	R^2
Linear Regression	190.0119	**282.9067**	**0.8424**
Random Forest	**184.0099**	309.8166	0.8110
Gradient Boosting	200.2995	290.2075	0.8342

Linear Regression has the lowest RMSE (€282.91) and the highest R^2 (0.8424). **Random Forest** has the lowest MAE (€184.01) but the highest RMSE (€309.82) and the lowest R^2 (0.8110), which suggests that it made some larger errors than the other models. **Gradient Boosting** falls between

the others on RMSE and R^2 and has the highest MAE (€200.30). No model is best on every metric. The differences are fairly small (for example, 0.0082 in R^2 between Linear Regression and Gradient Boosting) and come from one train-test split, so they should not be over-interpreted.

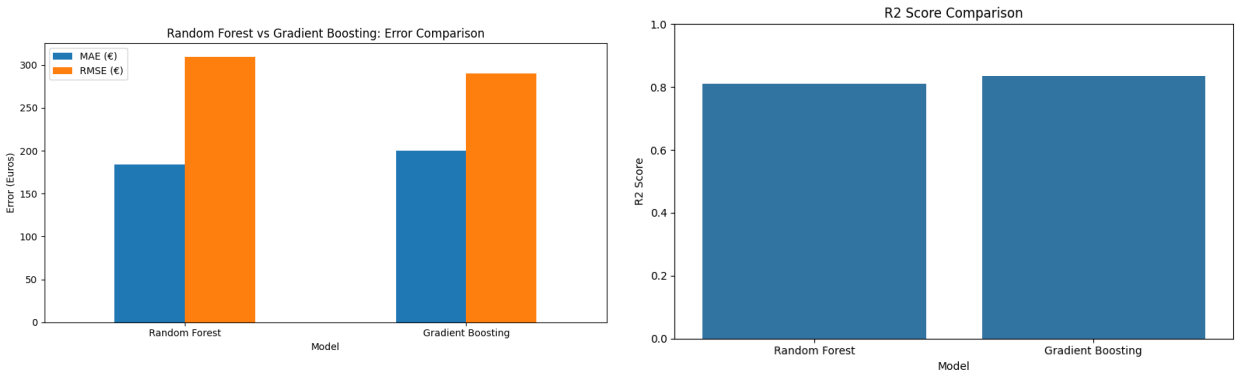


Figure 8. Random Forest and Gradient Boosting compared: (left) MAE and RMSE, (right) R^2 score. These are the notebook's two-model charts; Linear Regression was added later and is not shown in them.

9.3 Detailed error analysis: Gradient Boosting

The notebook examined the prediction errors of the Gradient Boosting model in detail. (A comparable error analysis was not carried out for Linear Regression or Random Forest.)

Table 13. Gradient Boosting: actual and predicted prices for the first ten test records.

Company	Product	Actual (€)	Predicted (€)	Abs. error (€)
Toshiba	Portege X30-D-10J	1,672.00	1,230.71	441.29
HP	Spectre 13-V100nv	1,149.00	1,223.99	74.99
Lenovo	IdeaPad 310-15ABR	499.00	563.26	64.26
Lenovo	IdeaPad Y700-15ISK	899.00	777.41	121.59
HP	EliteBook 820	1,244.00	1,002.78	241.22
HP	Spectre x360	1,398.99	1,442.40	43.41
HP	ProBook 430	719.00	716.83	2.17
Lenovo	IdeaPad 320-15ISK	459.00	727.09	268.09
HP	Omen 17-w212nv	1,191.00	1,235.85	44.85
HP	15-bw000nv (E2-9000e/4GB/500GB/Radeon	349.00	570.49	221.49

Table 14. Gradient Boosting: summary of absolute errors on the test set.

Statistic	Absolute error (€)
Count	261
Mean	200.30
Standard deviation	210.40
Minimum	0.31
25th percentile	65.24
Median	155.65
75th percentile	292.37
Maximum	1,737.75

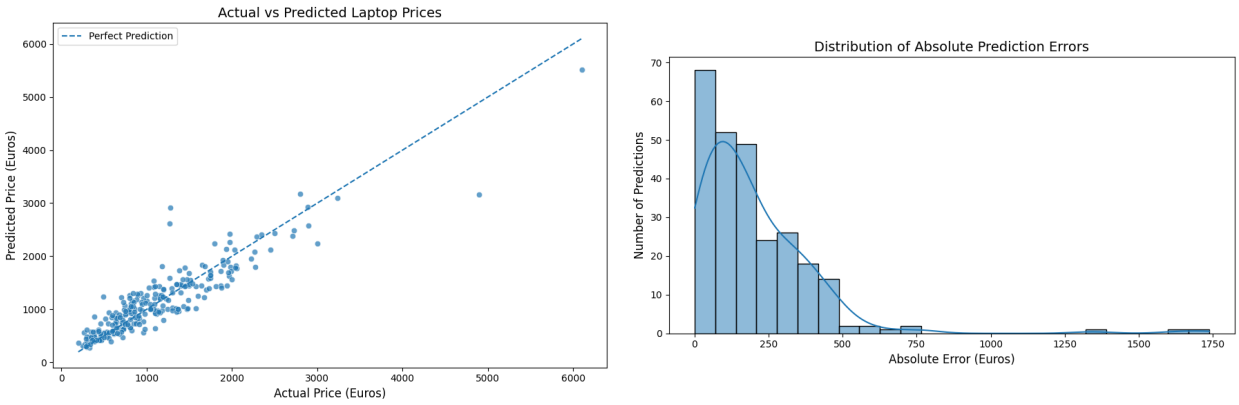


Figure 9. Gradient Boosting on the test set: (left) actual against predicted prices, with the dashed line marking perfect predictions; (right) distribution of absolute prediction errors.

Many predictions lie close to the perfect-prediction line, especially for lower- and middle-priced laptops, and half of the absolute errors are at or below about €155.65. A smaller number of errors are much larger, up to €1,737.75, and for some expensive laptops the model predicts a price well below the actual price. The first ten records show the same mix: the ProBook 430 was predicted within €2.17, while the Portege X30-D-10J was under-predicted by €441.29.

10. Classification Model Evaluation

10.1 Price categories

Table 15. Price categories used for classification.

Category	Definition
Low	Price ≤ €700
Medium	€700 < Price ≤ €1500
High	Price > €1500

The thresholds of €700 and €1500 are fixed, chosen by the project, and not learned from the data. Different thresholds could give different results.

10.2 Logistic Regression results

Logistic Regression was trained on the stratified 80% training split and evaluated on the 261 test laptops. The overall **accuracy was 0.8506** (85.06%), meaning that about 85 of every 100 test laptops were placed in the correct category.

Table 16. Logistic Regression classification report (test set).

Class	Precision	Recall	F1-score	Support
High	0.78	0.75	0.77	61
Low	0.93	0.94	0.93	84
Medium	0.83	0.84	0.83	116
Accuracy			**0.85**	261
Macro average	0.85	0.84	0.84	261
Weighted average	0.85	0.85	0.85	261

- **Precision**: of the laptops predicted to be in a category, the share that really belong to it.
- **Recall**: of the laptops that really belong to a category, the share the model found.

- **F1-score**: the harmonic mean of precision and recall, giving one balanced number per class.
- **Support**: the number of test laptops that truly belong to the class.
- **Macro average** treats every class equally; **weighted average** weights each class by its support.

Low-priced laptops are classified best (F1 = 0.93), followed by Medium (0.83). The High category is the hardest (F1 = 0.77, recall 0.75), so about a quarter of the High-priced laptops in the test set were not identified as High. The Medium class is the largest (116 laptops), and the High class the smallest (61).

10.3 Confusion matrix

A confusion matrix for the **Logistic Regression** classifier was not produced in the notebook, so it is not shown here. The notebook does contain a confusion matrix for a different analysis: the **Gradient Boosting regression** predictions and the actual test prices were each sorted into the same three categories, and the two sets of categories were compared. This is a supplementary view of the regression model, not the output of the Logistic Regression classifier.

Table 17. Confusion matrix: Gradient Boosting price predictions grouped into categories (261 test laptops).

Actual \ Predicted	Low	Medium	High	Total
Low	58	19	0	77
Medium	12	109	10	131
High	0	12	41	53

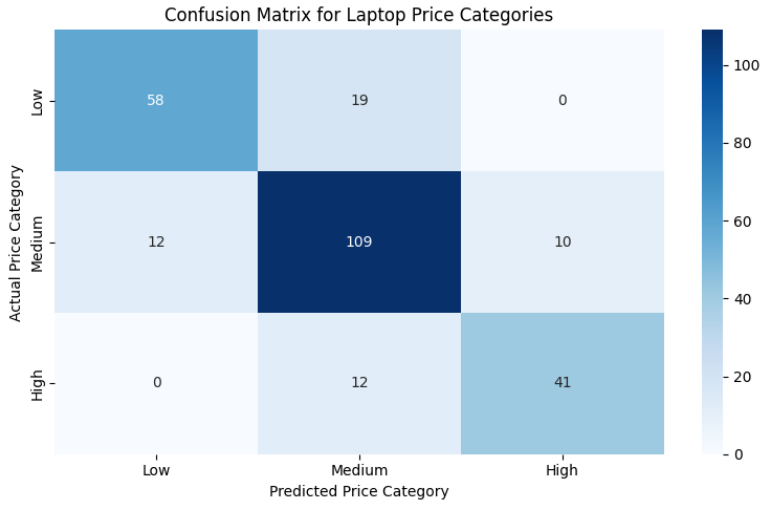


Figure 10. Confusion matrix for the Gradient Boosting predictions grouped into Low, Medium and High categories.

Table 18. Classification report for the Gradient Boosting predictions grouped into categories.

Class	Precision	Recall	F1-score	Support
Low	0.83	0.75	0.79	77
Medium	0.78	0.83	0.80	131
High	0.80	0.77	0.79	53
Accuracy			**0.80**	261
Macro average	0.80	0.79	0.79	261
Weighted average	0.80	0.80	0.80	261

Of the 261 test laptops, 208 (about 0.80) fall on the diagonal and are in the correct category. All errors are between neighbouring categories: no Low laptop was predicted as High and no High

laptop as Low. These results should **not** be compared directly with the Logistic Regression results. They come from a regression model whose output was grouped afterwards, and from a different, non-stratified test set (77 / 131 / 53 laptops per category, against 84 / 116 / 61 for Logistic Regression).

11. ROC Curve and ROC-AUC Analysis

A **ROC curve** shows how well a classifier separates one class from the rest as the decision threshold changes. It plots the true positive rate against the false positive rate. The **area under the curve (AUC)** summarises the curve in one number: 0.5 corresponds to random guessing and 1.0 to perfect separation. For a three-class problem, the notebook drew one curve per category, treating that category as “positive” and the other two as “negative” (one-vs-rest), using the predicted probabilities of the Logistic Regression model.

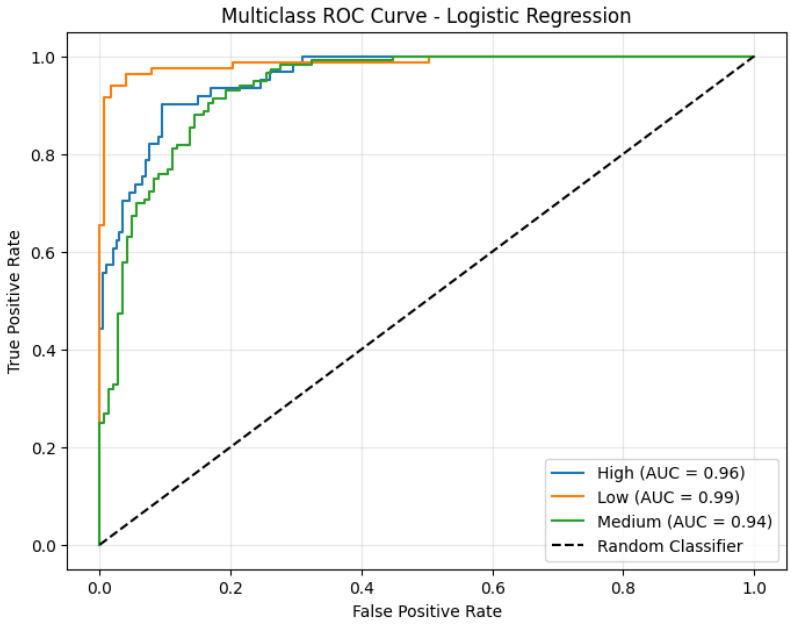


Figure 11. Multiclass ROC curves for the Logistic Regression model (one-vs-rest).

Table 19. Per-category ROC-AUC (as shown in the notebook’s ROC legend, to two decimals).

Category	ROC-AUC
High	0.96
Low	0.99
Medium	0.94

Table 20. Overall multiclass ROC-AUC for Logistic Regression.

Average	ROC-AUC	Meaning
Macro	0.9606	Simple average of the three category AUCs; each category counts equally.
Weighted	0.9581	Average of the category AUCs weighted by the number of test laptops in each category.

All three curves lie well above the diagonal random-classifier line. The Low category is separated best (AUC 0.99), then High (0.96) and Medium (0.94). The Medium category is the lowest, which is plausible because it sits between the other two categories, although the notebook does not test this explanation. The macro ROC-AUC (0.9606) and weighted ROC-AUC (0.9581) are close because the per-category values are similar. ROC-AUC does not depend on a single probability threshold, so it complements accuracy, precision, recall and F1-score.

12. Interpretation of Results

12.1 What the regression results mean

The three regression models have an average absolute error of roughly €184 to €200 on laptops whose median price is €977, and they explain between 81.10% and 84.24% of the variation in test prices. In practice, the models give a useful estimate of a laptop's price from its specifications, but an individual prediction can still be off by a few hundred euros, and occasionally by much more (the largest Gradient Boosting error was €1,737.75). The results should be treated as estimates.

That the simple Linear Regression model matches or beats the two ensemble models on RMSE and R^2 suggests that, with this dataset and these settings, the relationship between the encoded specifications and price is captured reasonably well by a linear model. The ensemble models were run with fixed settings and were not tuned, so the results do not show that Linear Regression would stay ahead of tuned ensembles.

12.2 Strengths and weaknesses shown by the results

- **Strengths:** R^2 values above 0.81 for every regression model; a classifier with accuracy 0.8506 and macro ROC-AUC 0.9606; consistent preprocessing in a single pipeline for every model; Low and Medium laptops classified reliably.
- **Weaknesses:** prediction errors remain, with some large ones for expensive laptops (in the Gradient Boosting analysis); High-priced laptops are the hardest class to classify (F1 = 0.77); the metrics rank the regression models differently, so no single model is best on all of them.

12.3 Classification results

Logistic Regression assigns a laptop to the correct price category about 85% of the time on the test set, and it ranks laptops by category very well (ROC-AUC of 0.94 to 0.99). It identifies Low-priced laptops best and High-priced laptops least well. This may help a buyer decide quickly whether a laptop is likely to be budget, mid-range or premium, but it gives no exact price.

13. Final Model Selection

13.1 Regression (price prediction)

Selected model: Linear Regression. The project's objective is to predict laptop prices numerically with small errors. Linear Regression has the lowest RMSE (€282.91) and the highest R^2 (0.8424), and it is also the simplest and easiest to explain of the three models.

Table 21. Trade-offs between the regression models (computed from the values in Table 12).

Comparison	Difference
Linear Regression vs Random Forest, MAE	Random Forest lower by €6.00 (184.01 vs 190.01)
Linear Regression vs Random Forest, RMSE	Linear Regression lower by €26.91 (282.91 vs 309.82)
Linear Regression vs Gradient Boosting, RMSE	Linear Regression lower by €7.30 (282.91 vs 290.21)
Linear Regression vs Gradient Boosting, R^2	Linear Regression higher by 0.0082 (0.8424 vs 0.8342)

Trade-offs. MAE treats all errors equally, while RMSE penalises large errors more. Random Forest has the lowest MAE, so it may be preferred when the typical (average) error matters most. Linear Regression has the lowest RMSE and the highest R^2, so it may be preferred when large errors are costly and when explaining more of the price variation is the goal. The gaps are small, they come from one train-test split, and the notebook did not run significance tests or cross-validation. This is

therefore a preference based on the current results, not a claim that Linear Regression is the best model for every dataset or for future data.

13.2 Classification (price category)

Selected model: Logistic Regression, the only classifier in the project. It reached an accuracy of 0.8506 and macro and weighted ROC-AUC values of 0.9606 and 0.9581, and it provides probability estimates for each category. Because regression and classification models answer different questions and use different metrics, the classifier is not ranked against the regression models. The two are chosen separately for their separate tasks.

14. Limitations

- **Dataset size and coverage:** 1,303 records may not represent every laptop model, manufacturer, market or price range, and some groups (for example 24 GB RAM or Workstation laptops) are very small.
- **Changing market prices:** prices change with new releases, discounts, inflation and demand. A model trained on historical prices may not match current prices. The notebook does not state when the prices were collected.
- **Evaluation method:** each model was evaluated once on a single 80/20 split. Results could change with another split, and cross-validation was not performed.
- **Prediction errors:** all regression models have non-zero errors, with some large errors for expensive laptops, so predictions are estimates, not guaranteed prices.
- **Fixed price thresholds:** the Low, Medium and High categories use fixed cut-offs of €700 and €1500. Other thresholds could change the classification results.
- **No external validation:** the models were not tested on a separate dataset of new laptops or current prices.
- **No tuning and limited preprocessing:** models were run with fixed settings and no hyperparameter search. Numerical features were not scaled, and the text columns (Cpu, Memory, Gpu, ScreenResolution) were one-hot encoded as raw text.
- **Product names as a feature:** Product (618 distinct names) was used as a categorical input, and the notebook did not test its effect.
- **Limited error analysis:** the detailed error analysis covers Gradient Boosting only, and a confusion matrix was not produced for the Logistic Regression model.

15. Future Improvements

The items below are **proposed future work**. None of them was carried out in the notebook.

- **Update the data:** collect a larger and more recent dataset with current prices and specifications.
- **Cross-validation:** evaluate every model across several train-validation splits to judge how stable the results are.
- **Hyperparameter tuning:** use Grid Search or Randomized Search with cross-validation, especially for Random Forest and Gradient Boosting.
- **Feature engineering:** derive features such as processor details, storage size and type, GPU category and screen resolution from the text columns, taking care to avoid information leakage. The effect of removing or grouping the Product feature could also be tested.
- **Model explainability:** use feature-importance or similar techniques to see which specifications influence predictions (bearing in mind that one-hot encoding splits one feature across many

columns).

- **External validation:** test the selected models on an independent dataset of new laptop models.
- **Web application:** build a simple application where a user enters specifications and receives an estimated price, a predicted category and a note about the limits of the prediction, using a validated model and the same preprocessing pipeline.

16. Final Conclusion

This project examined laptop prices using the Kaggle Laptop Price Dataset (1,303 laptops, 13 columns). The data was complete and free of duplicates; only **Ram** and **Weight** needed conversion to numbers. EDA showed a right-skewed price distribution, large differences in average price between brands and laptop types, a positive association between RAM and price (correlation 0.743), and almost no linear association between screen size and price (0.068).

Three regression models were evaluated on the same test set. Linear Regression had the lowest RMSE (€282.91) and the highest R^2 (0.8424), Random Forest had the lowest MAE (€184.01), and Gradient Boosting scored an MAE of €200.30, an RMSE of €290.21 and an R^2 of 0.8342. Linear Regression is the preferred regression model on RMSE and R^2, with Random Forest a strong alternative when MAE is the priority. For classification, Logistic Regression reached an accuracy of 0.8506, with F1-scores of 0.93 (Low), 0.83 (Medium) and 0.77 (High), a macro ROC-AUC of 0.9606 and a weighted ROC-AUC of 0.9581.

The main limitations are the single train-test split, the absence of cross-validation and tuning, historical prices, fixed category thresholds and the lack of external validation. The reported scores are estimates of performance on this dataset and not guarantees for new laptops. With updated data, cross-validation, tuning and external testing, the models could be made more reliable before real-world use.

17. References

[1] Laptop Price Dataset (Kaggle). Dataset used in this project. https://www.kaggle.com/datasets/muhammetvarl/laptop-price

[2] pandas documentation. https://pandas.pydata.org/docs/

[3] NumPy documentation. https://numpy.org/doc/

[4] Matplotlib documentation. https://matplotlib.org/stable/

[5] seaborn documentation. https://seaborn.pydata.org/

[6] scikit-learn documentation (user guide and API reference). https://scikit-learn.org/stable/

[7] scikit-learn: LinearRegression. https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html

[8] scikit-learn: RandomForestRegressor. https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html

[9] scikit-learn: GradientBoostingRegressor. https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.GradientBoostingRegressor.html

[10] scikit-learn: LogisticRegression. https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html

[11] scikit-learn: roc_auc_score. https://scikit-learn.org/stable/modules/generated/sklearn.metrics.roc_auc_score.html

[12] Google Colaboratory. https://colab.research.google.com/

Appendix A. Verification of Reported Results

Each reference value supplied for this report was checked against the saved outputs of the notebook. Every value matched, and no value was changed.

Table 22. Reference values against notebook outputs.

Item	Reference value	Notebook output	Status
Records / columns	1,303 / 13	(1303, 13)	Matches
Linear Regression	MAE 190.0119, RMSE 282.9067, R^2 0.8424	190.0119, 282.9067, 0.8424	Matches
Random Forest	MAE 184.0099, RMSE 309.8166, R^2 0.8110	184.0099, 309.8166, 0.8110	Matches
Gradient Boosting	MAE 200.2995, RMSE 290.2075, R^2 0.8342	200.2995, 290.2075, 0.8342	Matches
Logistic Regression accuracy	0.8506	0.8506	Matches
F1-scores (High / Low / Medium)	0.77 / 0.93 / 0.83	0.77 / 0.93 / 0.83	Matches
Macro / weighted ROC-AUC	0.9606 / 0.9581	0.9606 / 0.9581	Matches
Category thresholds	Low ≤ 700; Medium ≤ 1500; High > 1500	bins [0, 700, 1500, ∞]	Matches

Information not available in the notebook (and therefore not stated in this report):

- The number of laptops per brand, and the year or source of the price collection.
- Python and library version numbers.
- A confusion matrix for the Logistic Regression model.
- Unrounded per-category ROC-AUC values (the notebook shows them to two decimals).
- Any cross-validation, hyperparameter tuning, feature-importance analysis or external-data test.
- The Kaggle page’s own description of the dataset (the dataset details here come from the notebook).